23. DEVELOPMENT OF THE OESOPHAGUS

DURATION : 07:43 STUDY SHEET

COURSE SUMMARY

This video deeply explores the development of the oesophagus, highlighting its essential role as an articulation between the mind and the heart. It addresses the impact of embryonic flexion on the trunk, which shapes the digestive system and allows for the formation of the lungs and oesophagus. A focus is placed on the adaptive structure of the oesophagus, its connection to the diaphragm, and the consequences of dysfunctions, particularly regarding reflux and ENT problems. The dynamic interactions between the oesophagus, trachea, and aorta reveal a rhythmic complexity that is crucial for our overall health. Through key concepts such as the vagal plexus and peritoneal vasculature, this lesson illuminates the importance of the embryonic system for osteopathic practice.

OESOPHAGEAL DEVELOPMENT

The development of the **oesophagus** is an articulation between the **mind** and the **heart**. The **aorta** plays a central role in this system, while the **amniotic cavity** exerts external pressure. Between these two elements, the **brain** guides development and the **heart** serves as a fulcrum.

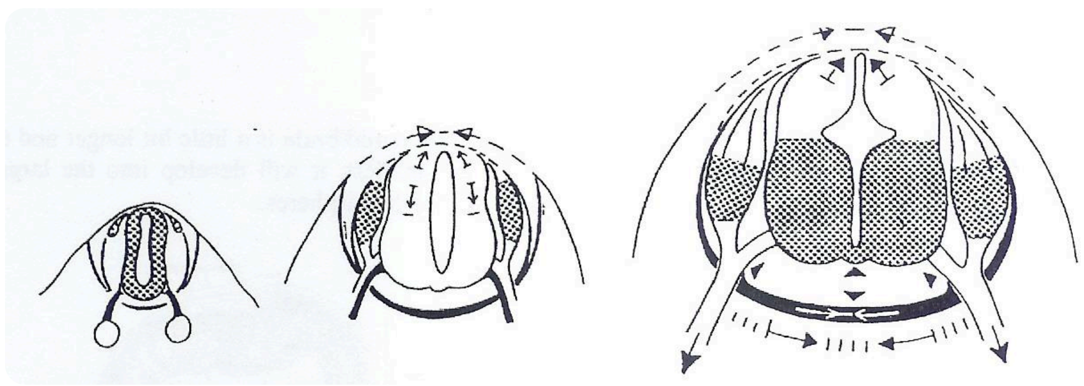

FIG. — Gastrulation neurulation invagination

We previously studied the influence of the applicature of the embryo's flexion movement at the head. Now, we will examine the influence of this applicature at the level of the **trunk**. This flexion movement generates various digestive phenomena.

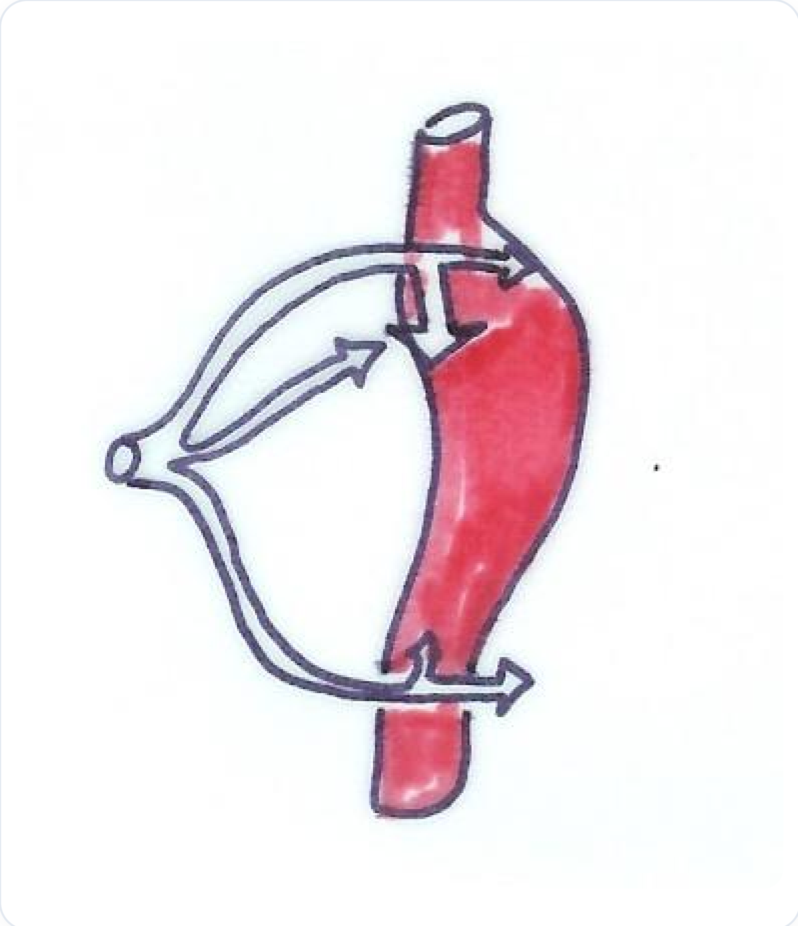

FIG. — Fetal circulation

One of the first digestive phenomena to study is the development of the **lungs**, which can be considered a gland. This flexion movement involves the **amniotic cavity**, the **vitelline cavity**, and the **heart**. We observe the appearance of the **foregut** with diverticula. Gradually, a pulmonary area and a digestive area take shape.

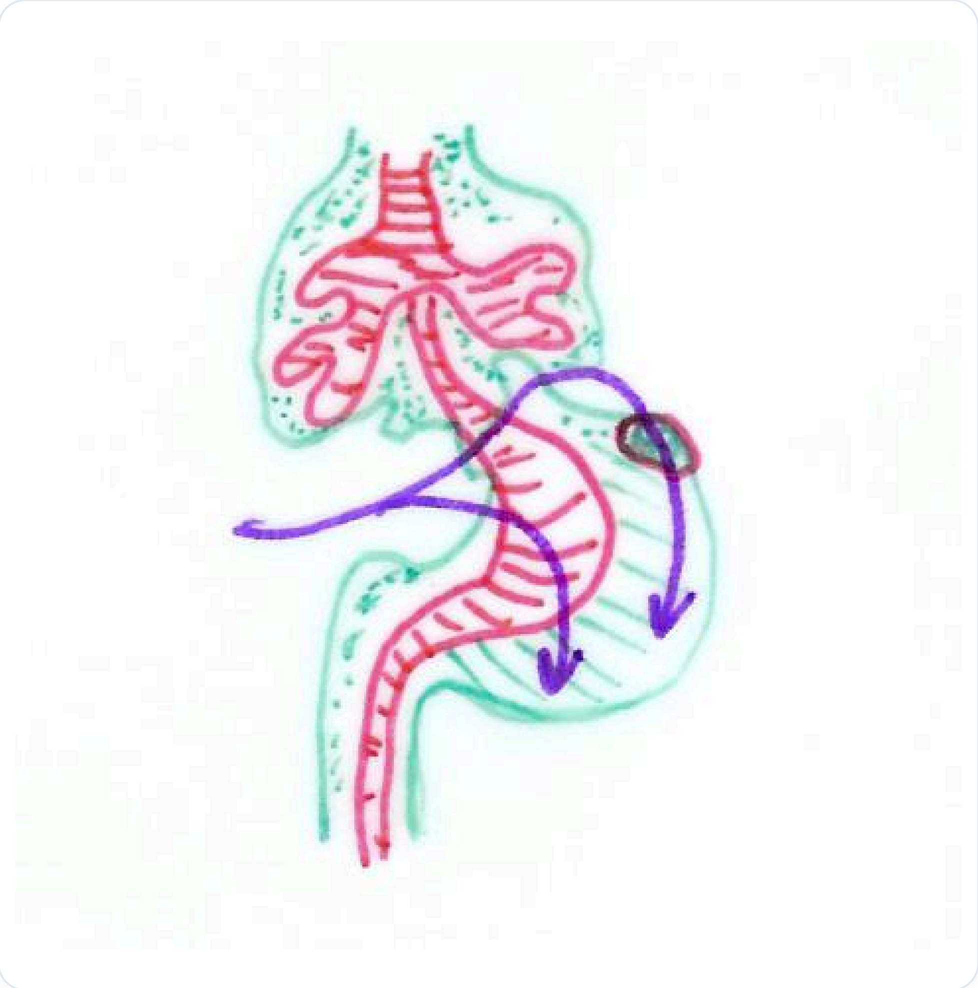

FIG. — Fetal circulation

The **oesophagus** develops from the segment of the **primitive gut**, located between the **respiratory diverticulum** and the dilatation of the **stomach**. It consists of an **endoblastic** part (internal tissue) and a **mesenchyme** (connective, muscular, and fibrous tissue). The epithelium forms the internal part, while surrounding it is tissue that contributes to its structure.

The **oesophagus** shows a certain adaptability to its environment. However, many problems at the oesophageal junction are related to its relationship with the **diaphragm**. These dysfunctions can lead to **ENT** problems, such as otitis, rhinitis, and even gastric reflux. Elements such as **pepsin** and **hydrochloric acid** have been found in the middle ear of some children, highlighting the link between reflux and ENT problems.

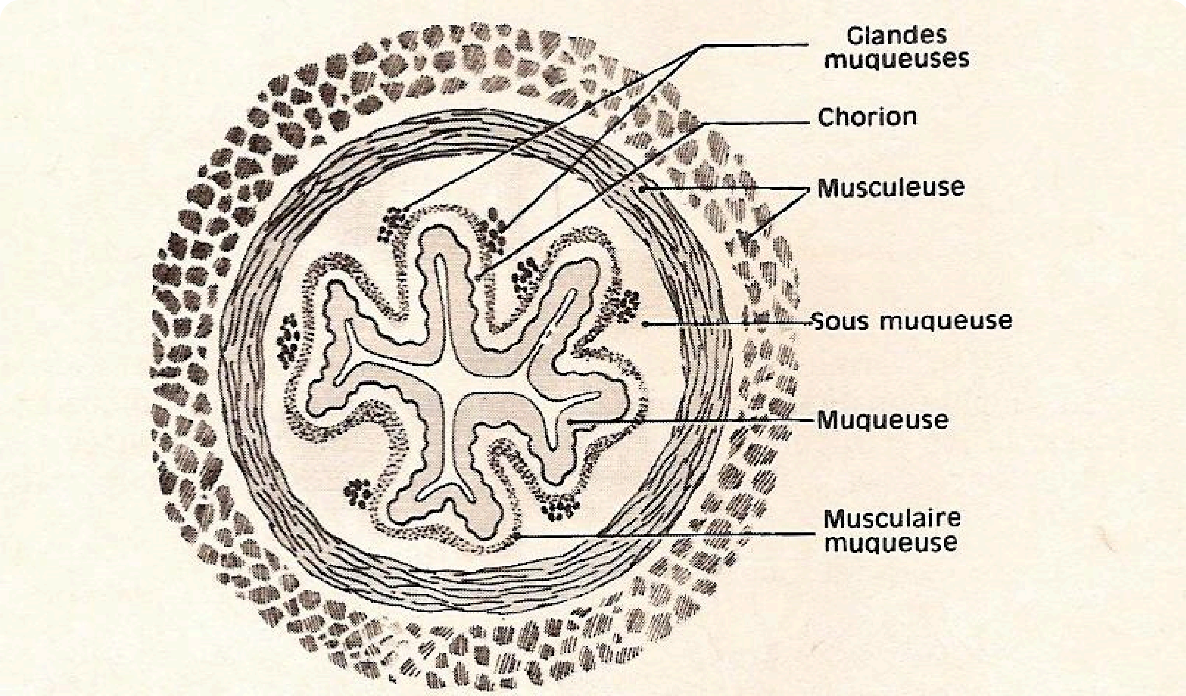

FIG. 53. — *Coupe transversale de l'œsophage.*

FIG. — Esophagus section

It is important to note that reflux does not always manifest as heartburn. Indeed, some refluxes are gaseous, involving gas molecules that rise into the system. This is often due to a dysfunction at the junction between the oesophagus and the stomach, at the diaphragmatic level.

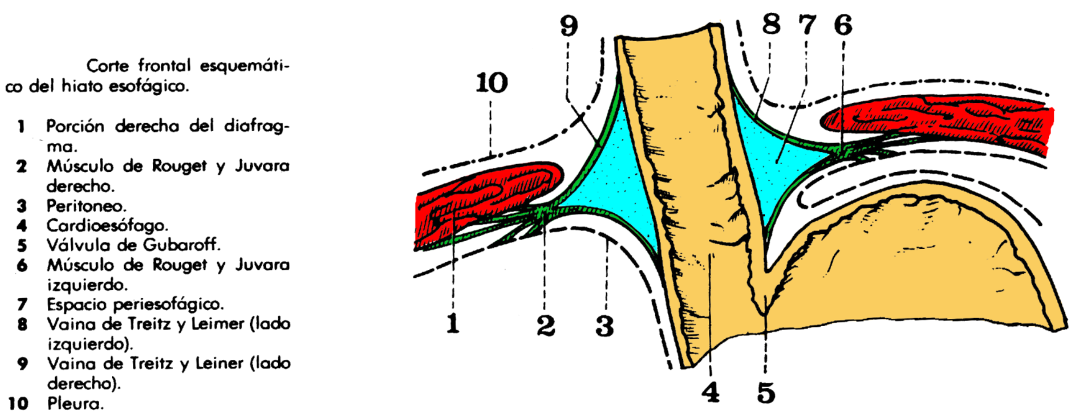

FIG. — Esophageal hiatus

The **diaphragmatic junction** is crucial because it is related to **Juvara's fibres** and the **peri-oesophageal space**, which is adaptive. A dysfunction at this level can block the quality and opening of the diaphragm. Steele said: "If your diaphragm is not free, the door to your life is not free."

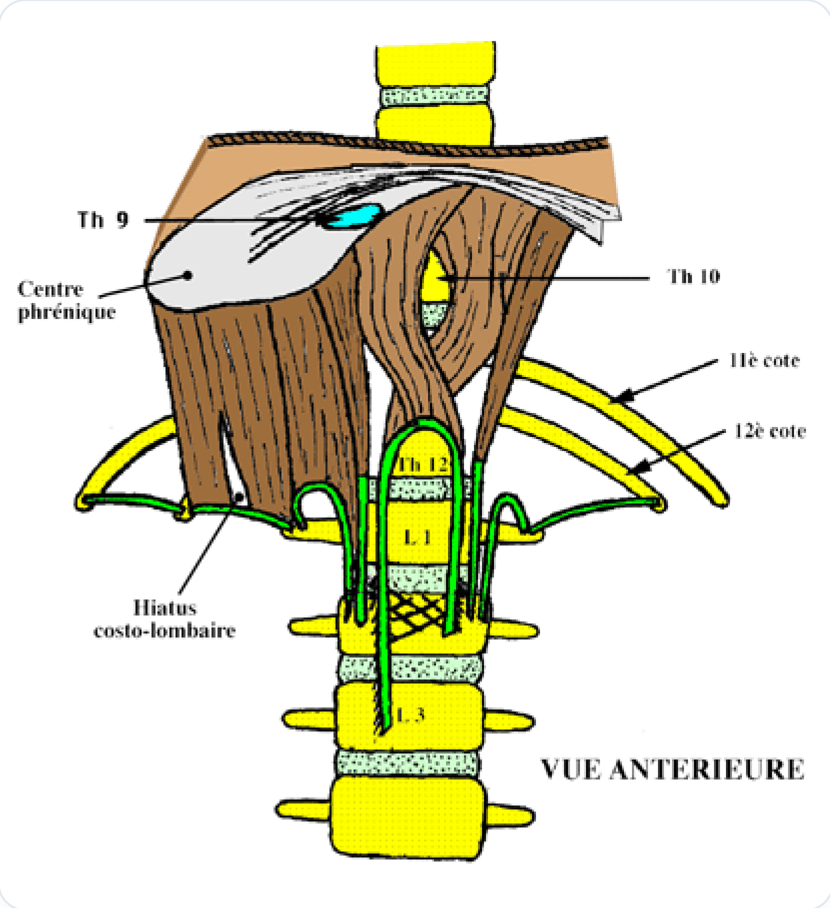

FIG. — Diaphragm anterior view

The **vagal plexus** is also very important. The **oesophagus**, **trachea**, and **aorta** are interconnected, forming a rhythmic zone. This zone is maintained by a **common fascia** and a **posterior mesentery**, which insert onto the mid-dorsal area, around T3-T4.

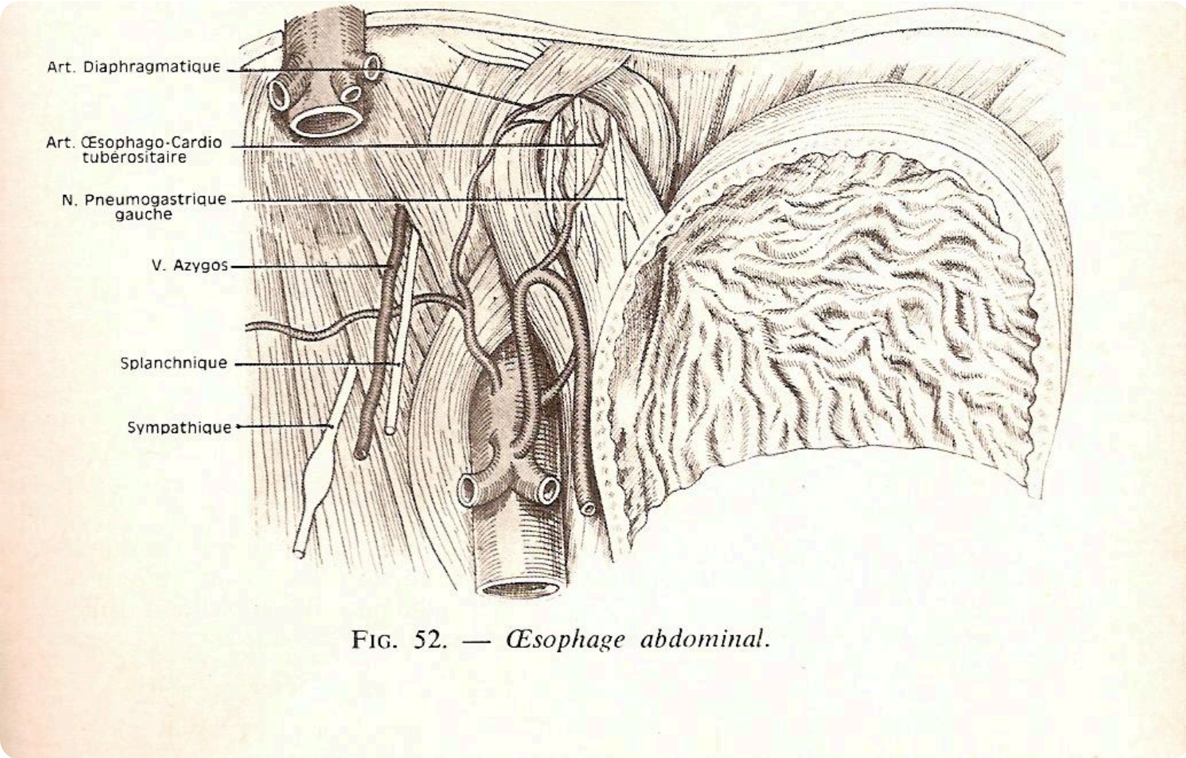

FIG. — Abdominal esophagus

The **oesophagus** is involved in swallowing, the **trachea** in breathing, and the **aorta** in the movement of the heart. These interactions occur thousands of times a day, creating a complex rhythmic dynamic.

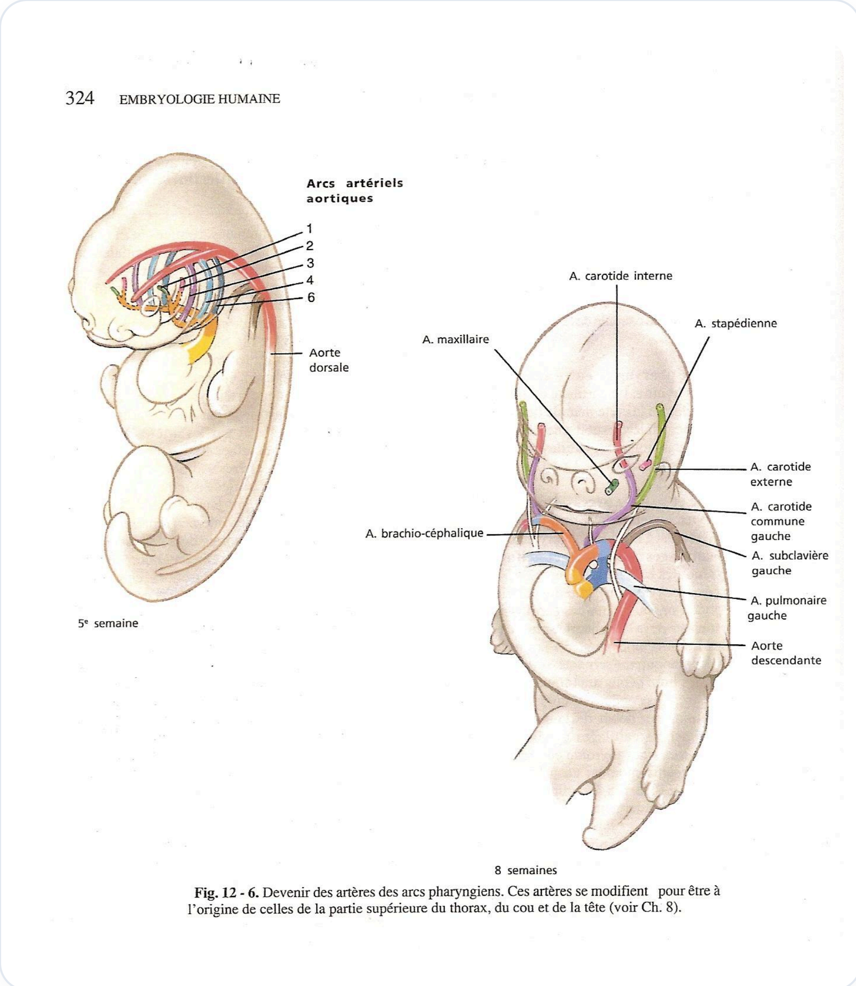

FIG. — Embryonic aortic arches

We also have a direct continuity with the aortic system, as well as a ligament that extends towards the **fourth part of the duodenum**, opening an important area at the level of the **coeliac trunk**. The latter is essential for the vascularisation of the sub-diaphragmatic peritoneal part, supplying the **liver** and the **stomach**.

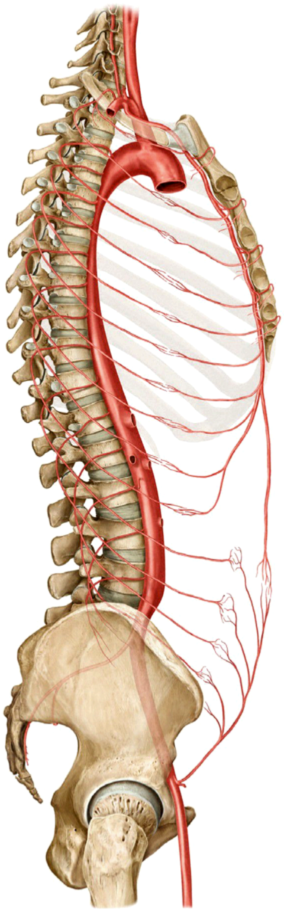

FIG. — *Aorta, arch, intercostals*

Finally, we will address the study of the **lungs**, reviewing the different layers of the oesophagus and their importance in the overall development of the digestive system.

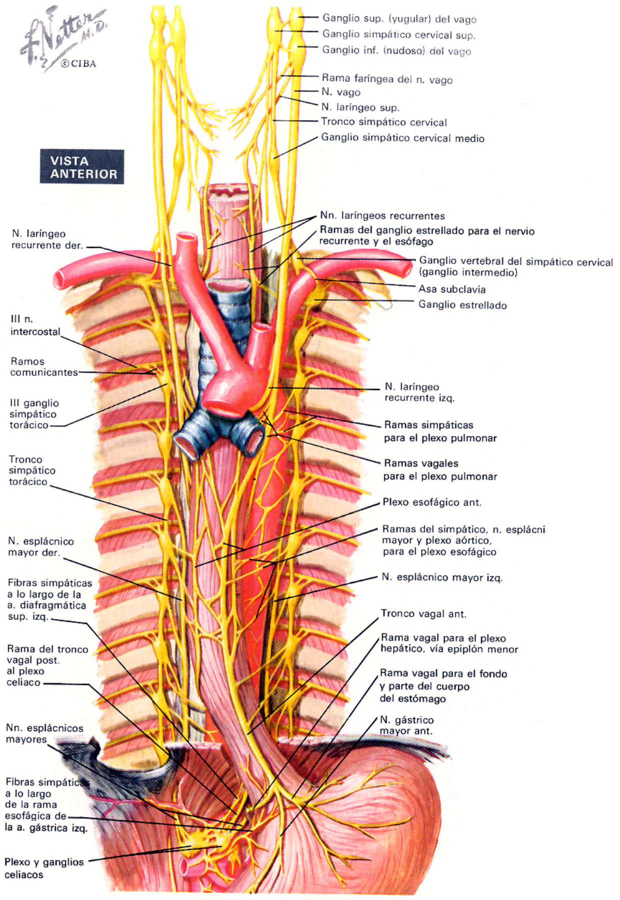

FIG. — Thoracic and abdominal innervation